

9. Access control

9.1. Configuration options

Lesson objectives

Access control: Configuration options

1. Explain the difference between authentication & authorization.
2. Describe different authentication options.
3. Describe different authorization options.
4. Explain the different options available for user management.

Security features and integrations

Secured communications

- Full TLS encryption support for BI/client tool connections
- TLS encryption within cluster nodes
- Specific data source connector security mechanisms
- Support for secure connections in client libraries including ODBC, JDBC, Python, and more

Authentication

- Password: LDAP / Active Directory, file-based, Salesforce
- Single sign-on: OAuth 2.0, Okta, AWS IAM, Azure IAM, Google Auth
- JWT
- Certificate
- Kerberos

Built-in access control

- Native, integrated RBAC
- Easily configure user data and platform access
- Flexible configuration of roles and privileges
- No need for 3rd party tool for security essentials

Additional access controls

- Immuta
- Privacera
- Apache Ranger
- File-based
- AWS Lake Formation

Secured connectors

- End-user impersonation
- Credential passthrough
- Secured authentication
- Encrypted data access and streaming

Auditing and monitoring

- Query and usage history
- Event logging
- Cluster monitoring
- Monitoring dashboards
- Audit logs for security changes



Two types of "auth"

Authentication

Verifies the **identity** of a user or service

Authorization

Determines the **access rights** of a user or service

Authentication options - Starburst Enterprise

Default

No default exists as the highly customizable Starburst Enterprise should be configured via the most appropriate option.



Supported

Multiple options, including the following:

- Password file
- LDAP
- Salesforce
- OAuth2
- Okta
- Kerberos
- Certificates
- Pass-thru delegated admin

Authentication options - Starburst Galaxy

Default

Password file (managed by the web UI)

Supported

Multiple options, including the following:

- Okta
- Azure AD
- Google Workspaces
- SAML

Authorization options - Starburst Enterprise

Default

No default exists as the highly customizable Starburst Enterprise should be configured via the most appropriate option.



Supported

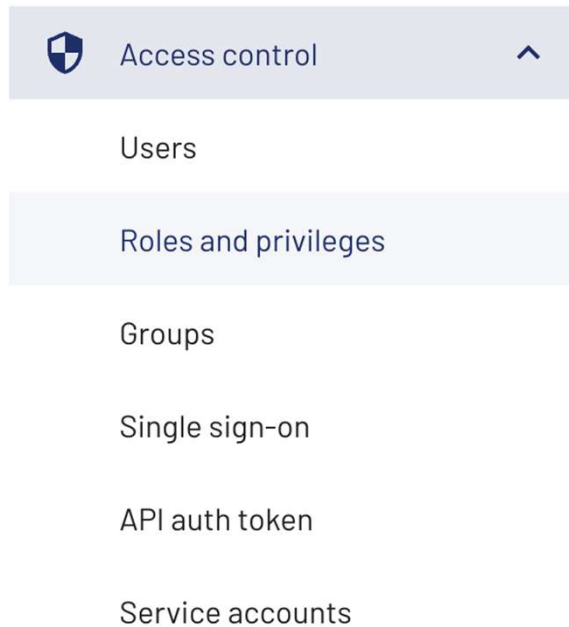
Multiple options, including the following.

- [Built-in access control](#) (BIAC)
- Apache Ranger
 - With Privacera
 - Global access control
 - Hive and Delta Lake control
- Immuta

Authorization options - Starburst Galaxy

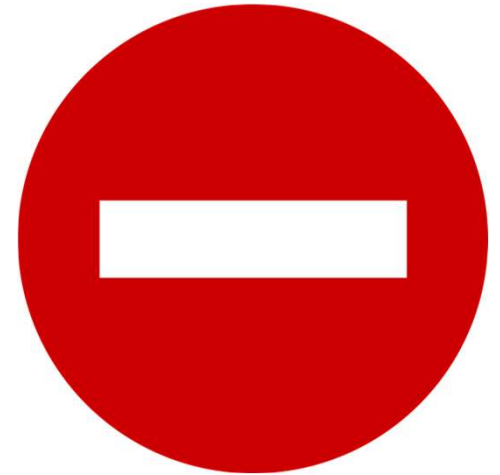
Default

[Built-in access control](#) (BIAC)



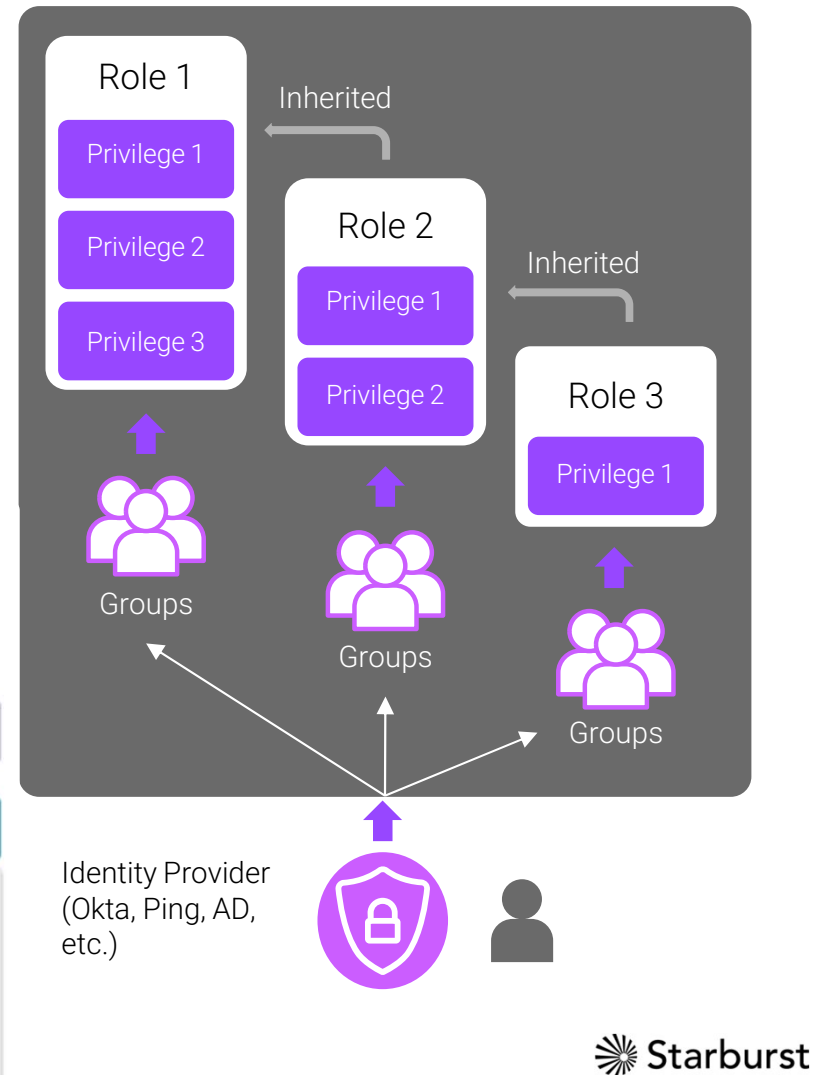
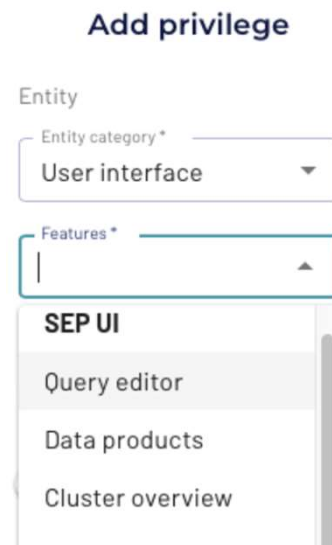
Supported

As of April, 2025, no other supported options exist



Built-in access control

- Integrated built-in role-based access control (RBAC) system with table, column, and row level access control
- Easy to configure any user's correct access rights to catalogs and to individual schemas, tables, and views
- Restrict or allow access to specific rows or columns
- Manage access for data products and UI screens
- Audit logs of all access control changes
- Row-level filtering and column-level masking



9. Access control

9.2. User management

Users

Starburst Galaxy

<> Query

Data

Jobs

Partner connect

Admin

Access

Authentication

API auth token

Groups

OAuth clients

Service accounts

Single sign-on

Users

Authorization

Column masks

Data classifier jobs

Roles and privileges

Row filters

Tags

Collapse

Users

Add user

2 users

Search users

Email	Roles	Last login	Created on
erin.rosas@starburstdata.com (you)	accountadmin, catalog_admin, catalog_admin_2, marketing	1 day ago	Aug 5, 2022
monica.miller@starburstdata.com	accountadmin	9 months ago	Dec 1, 2023

accountadmin

Users - Add user

The screenshot shows the Starburst Galaxy 'Users' management page. A modal window titled 'Add user' is open in the center. The modal contains three input fields: 'Email address *' (a text box with a cursor), 'Roles' (a dropdown menu showing 'public'), and 'Assign default role' (a dropdown menu showing 'public'). At the bottom of the modal are two buttons: 'Cancel' and 'Add user'. The background interface is dimmed and shows a sidebar with navigation options: Query, Data, Jobs, Partner connect, Admin, and Access. The main content area has a header 'Users' and a table with columns 'Email', 'Created on', and a search bar. The table lists two users: 'erin.rosas@start' and 'monica.miller@s'. The bottom right corner of the interface features a 'Collapse' button and a Starburst logo.

Starburst Galaxy

Query

Data

Jobs

Partner connect

Admin

Access

Users

Add user

Email ↑

erin.rosas@start

monica.miller@s

2 users

Search users

Created on

Aug 5, 2022

Dec 1, 2023

Add user

Cancel

Collapse

Starburst

Users - Edit user

Edit user

Erin Rosas | erin.rosas@starburstdata.com

Last login: Apr 4, 2025, 4:25 PM Created on: Aug 5, 2022, 3:55 PM User ID: u-3351546671

Your roles (4)

Inherited (3)

accountadmin

Default role



☒ Make this the default role

catalog_admin

catalog_admin_2

marketing

Assign a new role

Choose a role

▼

Assign

 Delete user

Groups

Starburst Galaxy

<> Query

Data

Jobs

Partner connect

Admin

Access

Authentication

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OAuth clients

Service accounts

Single sign-on

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Column masks

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Roles and privileges

Row filters

Tags

Collapse

Groups



SCIM has not been provisioned

Map group management with your identity provider, if available.

Provision SCIM

 accountadmin



Groups - Sync groups

Starburst Galaxy

<>

Query

Data

Jobs

Partner connect

Admin

Access

accountadmin

×

Configure single sign-on

Select an identity provider below and enter the values provided by your identity provider. Refer to your identity provider's SSO setup documentation for help finding this information.

[Learn how to configure SSO for Starburst Galaxy](#)

Select identity provider

☒ Okta

☐ Microsoft Entra ID

☐ Google Workspace

☐ Custom IdP

Okta configuration

Single sign on URL

https://erosas.galaxy.starburst.io/saml/v2/pn3Ep#fhXpP/acs

Audience URI (SP Entity ID)

https://erosas.galaxy.starburst.io/saml/v2/pn3Ep#fhXpP/metadata

Default RelayState

ewogICJyZWVpcnVjdFhhdG9lIDogIi81LAogICJvYXV8aCIgOiBmYWxzZQp9

Configure Starburst Galaxy

☒ Metadata URL

☐ Manual entry

Identity Provider metadata URL *

Configure single sign-on

Collapse

Lesson summary

Access control: Configuration options

1. Authentication verifies the identity of a user or service.
2. Authorization determines the access rights of a user or service.
3. Both Starburst Galaxy and Starburst Enterprise include a number of different options of authentication.
4. Both Starburst Galaxy and Starburst Enterprise include a number of different options of authorization.
5. Users can be managed in a number of ways, including: adding, editing, grouping, and syncing them.

9. Access control

9.3. Role-based access control

Lesson objectives

Access control: Role-based access control

1. Describe what Role-based access control (RBAC) is and how it is used.
2. Explain which privileges exist, and describe their use cases.
3. Demonstrate how RBAC policies operate in Starburst Galaxy.

Key concepts of role-based access control

Role-based access control (RBAC) assigns access entity privileges to roles, which are then assigned to users and groups

- **Entity** - a resource that needs to be secured
 - Examples: cluster, catalog, table, function, UI elements
- **Privilege** - authorization to perform a particular action against an entity
 - Examples: start/stop a cluster, read from a schema, insert into a table
- **Role** - defined set of privileges
 - Roles can contain roles
- **User** - a human or service account
- **Group** - defined set of users

Roles

Starburst Galaxy

<> Query

Data

Jobs

Partner connect

Admin

Access

Roles

Add role

7 roles

Search roles

Role name	Description	Granted to roles	Roles granted to	Owning role	Created on	
accountadmin	-	_system	role_testing_api	_system	Aug 5, 2022	🔒
catalog_admin	a role with typical catalog admin privileges			accountadmin	Jul 19, 2023	⋮
catalog_admin_2	-			accountadmin	Jul 20, 2023	⋮
deny_pii	deny access to all personally identifiable info...	marketing		accountadmin	Jul 8, 2024	⋮
marketing	-		deny_pii	accountadmin	Jul 8, 2024	⋮
public	-	_system		_system	Aug 5, 2022	🔒
role_testing_api	Role created by Postman to test an API endp...	accountadmin		accountadmin	Oct 2, 2023	⋮

Authentication

API auth token

Groups

OAuth clients

Service accounts

Single sign-on

Users

Authorization

Column masks

Data classifier jobs

Roles and privileges

Row filters

Tags

Collapse

Roles - Add role

The screenshot displays the Starburst Galaxy Roles management interface. A modal window titled "Add a new role" is open, allowing users to create a new role. The modal includes a "Role name" input field, a "Description" text area, and a checkbox labeled "Grant to the creating role?". An "Add role" button is located at the bottom right of the modal. In the background, the "Roles" page is visible, featuring a sidebar with navigation options (Query, Data, Jobs, Partner connect, Admin, Access) and a table of existing roles.

Role name	Created on
accountadmin	Aug 5, 2022
catalog_admin	Jul 19, 2023
catalog_admin_2	Jul 20, 2023
deny_pii	Jul 8, 2024
marketing	Jul 8, 2024
public	Aug 5, 2022
role_testing_api	Oct 2, 2023

Roles - Edit role

Starburst Galaxy

Query

Data

Jobs

Partner connect

Admin

Access

accountadmin

Roles

Back

Catalog_admin role

a role with typical catalog admin privileges

Users

Roles

Groups

Privileges

Policies

Owned entities

Add privilege

Only directly assigned privileges

Allow

Allow and grant to others

Deny

Search privileges

Account	5 privileges	
Clusters	8 privileges	
Catalogs	21 privileges	
Locations	1 privilege	

Privileges

Add privileges to catalog_admin role

Allow this role to access all or specific catalogs, tables, or clusters within your organization. Refer to our detailed [documentation on assigning privileges to roles](#).

Data Account Cluster Location Function Data product

What would you like to modify privileges for?

Scope
glue

Do you want to allow or deny access?

☒ Allow ☐ Deny

Do you want to allow this role to grant to others?

☐ Allow role receiving privilege to grant to others

What can they do?

Privileges for this entity are now staged for granting

Privileges added

Data 7

glue

Allow: Manage table data observability X

Allow: Delete from table X

Allow: Update table rows X

Allow: Insert into table X

Allow: Select from table X

Allow: Create schema X

Allow: Create table X

Account

Cluster

Cancel

Save privileges (7)

Hands-on exercise

Creating and validating RBAC policies
with Starburst Galaxy (30 mins)

Lesson summary

Access control: Role-based access control

1. Role-based access control (RBAC) assigns access entity privileges to roles, which are then assigned to users and groups.
2. Roles vary by organization, but typically reflect business functions.
3. Admin users can add roles, edit them, grant privileges, or restrict privileges as needed.

9. Access control

9.4. Attribute-based access control

Lesson objectives

Access control: Attribute-based access control

1. Describe what Attribute-based access control (ABAC) is and how it is used.
2. Explain Tag-based access control (TagBAC) and how it relates to ABAC.
3. Explain how RBAC and ABAC work together.
4. Demonstrate how tag-based policies can be created and tested.
5. Demonstrate how row filters operate in Starburst Galaxy.

Key concepts of attribute-based access control

Attribute-based access control (ABAC) allows the combination of policies and attributes, such as tags, to manage role access to entities.

- Existing RBAC concepts – Entity, Privilege, Role, User, & Group
- Attribute - a wide range of characteristics (initially Tags in Galaxy)
 - Examples: tags, user status, citizenship, time of day, IP address, device type
- Policy - combination of attributes that are evaluated against 1 or more entities to allow/deny 1 or more privileges on the entities
 - Example
 - Entities: All schemas within the students catalog
 - Attributes: A tag named pii and/or any nested tags are present
 - Privileges: Deny reading any affected tables and/or columns
 - Policies are elements of a role
 - Roles can have zero to many policies

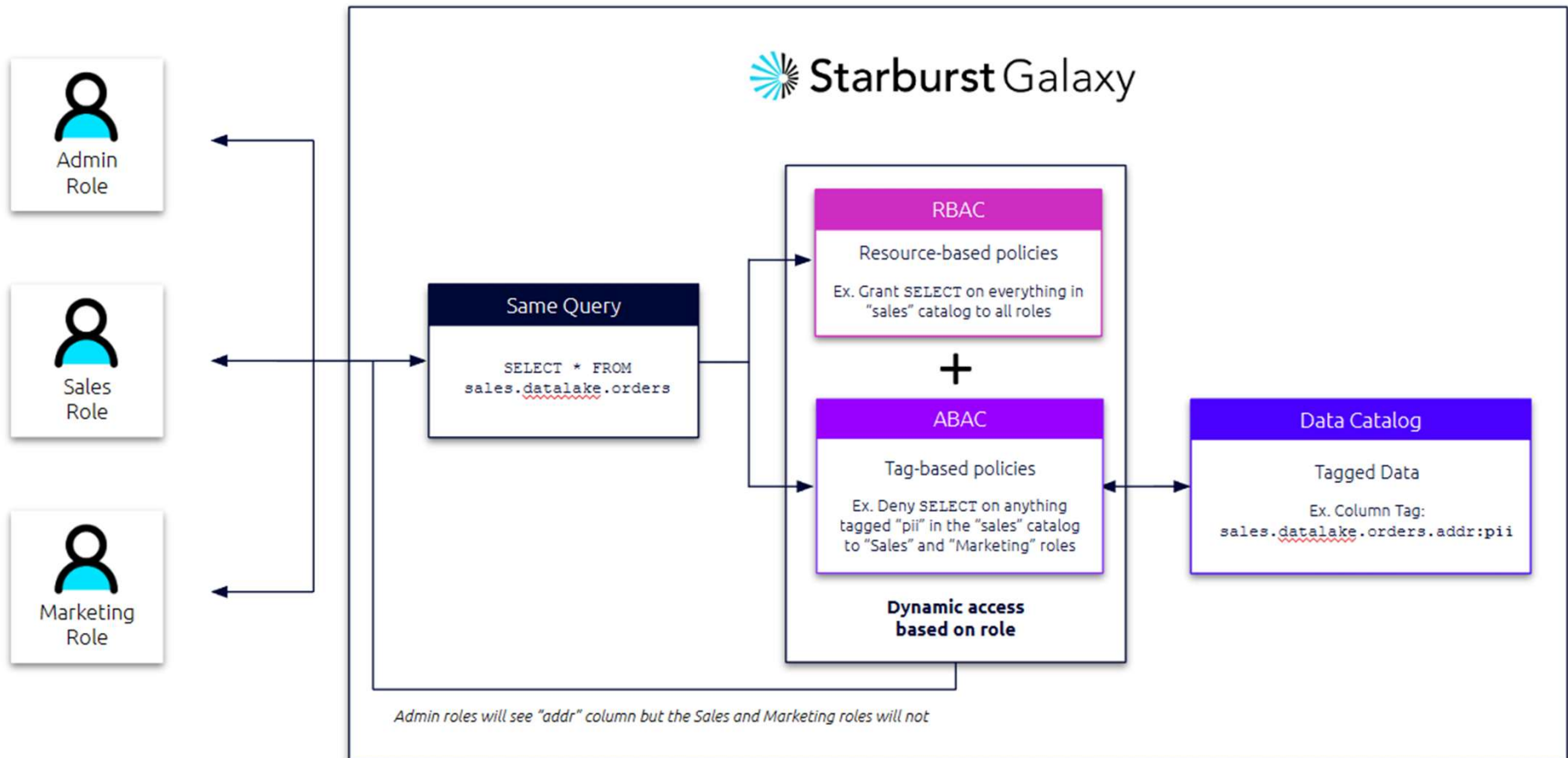
Tag-based access control

Tag-based access control (TagBAC) is the initial ABAC functionality available in Starburst Galaxy.

- An access control model that uses tags or labels to define access policies.
- Tags are associated with entities.
- Access decisions are based on the presence or absence of matching tags.

TagBAC can be considered a subset or implementation of ABAC, as the use of tags or labels can be one aspect of an ABAC system.

RBAC & ABAC work together



Create a tag

Starburst Galaxy

<> Query

Data

Jobs

Partner connect

Admin

Access

Authentication

API auth token

Groups

OAuth clients

Service accounts

Single sign-on

Users

Authorization

Column masks

Data classifier jobs

Roles and privileges

Row filters

Tags

Collapse

Tags

Create tag

6 tags

Search tags

missions

pii

Tag name	In-use	Comment	Owning role	Created	Last updated
missions	0	Identifies mission information	accountadmin	Dec 14, 2023, 4:31 PM	Dec 14, 2023, 4:31 PM
pii	1	personally identifiable information	accountadmin	Dec 18, 2023, 5:01 PM	Dec 18, 2023, 5:01 PM

?

Associate tags to entities

Catalogs

>

orders

Last indexed: Mar 28, 2025, 9:24:46 PM GMT-4

Description

Source comment

Default Type

Tags

Contacts

Owner

-

No source comment...

Iceberg

0

0

accountadmin

Columns 8

Metrics

Quality

Lineage

Definition

Data preview

Query history

Audit log

Access control

Refresh

Auto tag

8 columns

Search columns

Column ↑	Index status	Type	Nullable	Default	Tags		Description
customerid		integer	yes	NULL	pii		No description provided.
invoicedate		varchar	yes	NULL	No tags assigned.		No description provided.
invoiceno		integer	yes	NULL	No tags assigned.		No description provided.
itemtype		varchar	yes	NULL	No tags assigned.		No description provided.
quantity		integer	yes	NULL	No tags assigned.		No description provided.
salesid		integer	yes	NULL	No tags assigned.		No description provided.

Create a policy with a new/existing role

Starburst Galaxy

<> Query

Data

Jobs

Partner connect

Admin

Access

Roles

Catalog_admin role

a role with typical catalog admin privileges

Users

Roles

Groups

Privileges

Policies

Owned entities

Add policy

1 policy

Search policies

Policy name	Policy expression	Description	Scope	Role	Expires on	Created on
deny_pii_2	true	-	edu_demo.discovered_...	catalog_admin	Never	Apr 4, 2025, 4:59 PM

Policy - Name it

Define policy details

Allow this role to access all or specific catalogs, schemas, or tables within your organization. For more information, see [policies documentation](#)

Policy name and description

Policy name *

deny_pii_2

Can only contain lowercase letters (a-z), numbers (0-9), and underscores (" _")

Description

Policy expiration date and time (optional)

☐ Apply an expiration date to this policy?

Expiration date & time

DD MMMM YYYY hh:mm aa

- 1 Define policy details
- 2 Define policy scope
- 3 Add privileges, column masks, and row filters

Cancel

Next

Policy - Align it to entities

Scope

The objects selected limit what the matching expression can apply to.

Catalogs *

edu_de... ×



Schemas *

edu_demo.discovered_sche... ×



Tables *

edu_demo.discovered_schema.invoice_total_c... ×



Policy - Define an expression

Limit the scope

- ☐ Match everything in scope
- ☒ Match using expression 

```
has_tag(pii) OR has_tag(pii.*)
```

Policy - ALLOW/DENY privileges

Add privileges, column masks, and row filters

At least one of the following must be applied.

Privileges

What type of access should be allowed or denied?

Grant *

DENY



Privileges *

Select



+ Add privilege

Associate the role

Roles

< Back

Catalog_admin role

a role with typical catalog admin privileges

Users

Roles

Groups

Privileges

Policies

Owned entities

Assign role

1 role

Assigned role name ↑

deny_pii

⊖ Revoke

Test it out!

65	<code>SELECT name, phone, address FROM customer;</code>		
66			

 Failed	Avg. read speed	Elapsed time	Rows
	-	0.38s	-



Access Denied: Cannot select from columns [address, phone, name] in table or view students.yourname_sales.customer: Role mktg does not have the privilege SELECT on the columns [address, phone]

Additional feature: Row-level filtering

Allows data admins to filter tables dynamically based on a user's role, instead of utilizing these legacy strategies.

- Creating and storing individual subsets of the data
- Creating and securing table views

For example, showing only orders for a particular country code

** Implemented as SQL syntax thus needs to be tested

Create a row filter

Create row filter

×

Name *

regular_orders

16 characters remaining

Description

Regular orders only. Currently this is where status is F or O only.

432 characters remaining

Filter expression *

orderstatus IN ('F', 'O')

An SQL expression that will be appended to queries on tables in the scope

Cancel

Create row filter

Add it to a new/existing policy

Edit policy

[< Back](#)

Row filters

Create or add an existing row filter to apply to the above scope.

Row filter *	Filter expression	
regular_orders ▼	orderstatus IN ('F', 'O')	 

[+ Add row filter](#)

Test it out!

Without filter (students role)

TagBAC x +

Run selected (limit 1000) aws-us-east-1-free students

```
71 SELECT orderstatus, count() nbr_orders
72 FROM orders
73 GROUP BY orderstatus
74 ORDER BY orderstatus;
75
```

Finished Avg. read speed 14.8K rows/s Elapsed time 1s Rows 3

orderstatus	nbr_orders
F	7304
O	7333
P	363

With filter (mktg role)

TagBAC x +

Run selected (limit 1000) aws-us-east-1-free students

```
71 SELECT orderstatus, count() nbr_orders
72 FROM orders
73 GROUP BY orderstatus
74 ORDER BY orderstatus;
75
```

Finished Avg. read speed 13.7K rows/s Elapsed time 1s Rows 2

orderstatus	nbr_orders
F	7304
O	7333

Hands-on exercise

Creating and validating ABAC policies
with Starburst Galaxy (30 mins)

Lesson summary

Access control: Attribute-based access control

1. Attribute-based access control (ABAC) regulates entity access based on attributes combined into a policy linked to roles, which are then assigned to users & groups.
2. Tag-based access control (TagBAC) is a subset of ABAC which is focused on labeling entities with tags and then basing access decisions on the presence/absence of matching tags.
3. RBAC and ABAC work together to determine the most restrictive access controls to utilize.
4. Starburst web UI allows tags-based policies to be created and tested.
5. Row-level filters can eliminate the need to create separate tables or views.

10. Data pipelines

10.1. Definition & differentiation

Lesson objectives

Data pipelines: Definition & differentiation

1. Define a data pipeline and its typical use case.
2. Differentiate between ETL and ELT.
3. Explain how Change Data Capture (CDC) impacts data pipelines.
4. Explain how Slowly Changing Dimensions (SCD) impact data pipelines.